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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 58.8

$$\begin{aligned}
 & \int \frac{1}{4 \cos x + 3 \sin x} dx \\
 \text{t-formules: } t = \tan \frac{x}{2} & \Rightarrow dt = \frac{1}{2} \left(1 + \tan^2 \frac{x}{2} \right) dx \Leftrightarrow dx = \frac{2}{t^2 + 1} dt \\
 & = \int \frac{1}{4 \frac{1-t^2}{1+t^2} + 3 \frac{2t}{1+t^2}} \cdot \frac{2}{t^2 + 1} dt \\
 & = \int \frac{1+t^2}{4-4t^2+6t} \cdot \frac{2}{t^2 + 1} dt \\
 & = 2 \int \frac{1}{4-4t^2+6t} dt \\
 & = \int \frac{1}{-2t^2+3t+2} dt \\
 & -2t^2+3t+2=0 \Leftrightarrow (t-2)(-2t-1)=0 \\
 & \Rightarrow \frac{1}{-2t^2+3t+2} = \frac{A}{t-2} + \frac{B}{-2t-1} \\
 A = \left(\frac{1}{-2t-1} \right) \Big|_{x=2} & = -\frac{1}{5} \\
 B = \left(\frac{1}{t-2} \right) \Big|_{x=-\frac{1}{2}} & = -\frac{2}{5} \\
 & \Rightarrow \int \frac{1}{-2t^2+3t+2} dt = -\frac{1}{5} \int \frac{1}{t-2} dt - \frac{2}{5} \int \frac{1}{-2t-1} dt \\
 & = -\frac{1}{5} \ln |t-2| + \frac{2}{10} \ln |-2t-1| + c \\
 & = -\frac{1}{5} \ln \left| \tan \frac{x}{2} - 2 \right| + \frac{1}{5} \ln \left| -2 \tan \frac{x}{2} - 1 \right| + c
 \end{aligned}$$

References